Simulation of wastewater treatment plant

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***Abstract*—This paper presents the development and validation of a custom computational engine for the simulation of municipal wastewater treatment plants. Due to current infrastructure damage at the centralized Khartoum municipal facilities, the design basis was adapted to model a decentralized, neighbourhood-scale treatment plant. Standardized generation metrics characteristic of water-stressed urban districts was utilized to establish reliable influent baselines. The simulation focuses on the secondary biological treatment phase, employing a first-order Continuous Stirred-Tank Reactor (CSTR) mathematical model to calculate Biochemical Oxygen Demand (BOD) removal and parallel unit sizing requirements. Finally, the custom algorithmic logic was successfully validated against industry-standard commercial software (SuperPro Designer), demonstrating high fidelity in predicting effluent concentrations and required reactor volumes for capacity expansion.**

***Keywords*—Wastewater treatment, continuous stirred-tank reactor (CSTR), computer simulation, biochemical oxygen demand (BOD), decentralized treatment, SuperPro Designer.**

I. Introduction

The treatment of wastewater is a critical component of modern industrial and municipal infrastructure, driven by stringent environmental regulations and the global necessity for sustainable water management. Designing an efficient wastewater treatment plant (WWTP) requires a deep understanding of the complex physical, chemical, and biological processes involved in removing contaminants. In chemical engineering, process simulation provides a powerful, cost-effective method to model these unit operations, predict plant behaviour under varying conditions, and optimize the overall system design before physical implementation.

Background and Motivation

The complexity of wastewater streams often containing a highly variable mixture of suspended solids, organic matter, and chemical pollutants makes the design and scaling of treatment units challenging. To ensure that effluent meets environmental discharge standards, precise mathematical modelling of each unit operation is required.

Due to the unavailability of real-time municipal flow data resulting from recent infrastructural disruptions, the design basis for this simulation is established using standardized per-capita generation metrics applied to a localized population scale in Khartoum. The volumetric flow rate (Q) is estimated based on the population equivalent (PE) and standard per-capita wastewater generation rates, while the influent concentrations for biochemical oxygen demand (BOD) and chemical oxygen demand (COD) are derived from benchmark municipal data in established chemical engineering literature.

This approach serves a dual purpose: it validates the mathematical model against recognized theoretical standards while creating a scalable, flexible simulation framework. This foundational model can readily integrate empirical data once available, making it a highly relevant tool for future post-conflict infrastructure rehabilitation and water management planning. This project addresses the need for a rigorous, mathematically grounded simulation of a wastewater treatment facility to evaluate its operational efficiency under these designated baseline parameters.

Project Objectives

The primary objective of this project is to develop a comprehensive simulation of a wastewater treatment plant. To achieve this, the following key goals are established:

- **Process Modelling:** To establish accurate mathematical models for the key unit operations within the plant.
- **Mass Balance and Derivations:** To formulate detailed step by step mass balance equations for the system, moving from fundamental accumulation principles to specific design equations.
- **Simulation and Analysis:** To simulate the steady state operation of the plant and analyse the performance of individual units.
- **Performance Comparison:** To compare the simulated results with theoretical expectations and evaluate the system's efficiency under specific operating parameters.

Scope of the Report

This report details the foundational steps of the plant design. It begins with a process description, followed by the explicit derivation of the governing equations and mass balances for the treatment units. Finally, the results of the simulation are presented and discussed. This foundational modelling establishes the framework required for future project phases, including comprehensive energy balances across all units and economic evaluations.

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II. PROCESS DESCRIPTION

The proposed wastewater treatment plant operates through a series of sequential unit operations designed to progressively reduce the biochemical oxygen demand (BOD) and total suspended solids of the influent stream. The overall conceptual process is broadly categorized into primary, secondary, and sludge treatment phases.
A high-level overview of this configuration is illustrated in Figure 1.

A. Primary Treatment

The influent wastewater first undergoes primary treatment, which primarily relies on physical separation mechanisms. The stream enters the primary clarifier (sedimentation tank). In this unit, gravitational settling removes a significant portion of the heavier, settleable suspended solids and their associated organic matter. The underflow from this unit consists of primary sludge, while the clarified liquid (primary effluent) is directed to the secondary treatment phase.

B. Secondary (Biological) Treatment

The core of the biochemical conversion in the plant occurs during secondary treatment, which utilizes an activated sludge process. The primary effluent enters the aeration tank (bioreactor), where it is mixed with a culture of microorganisms and supplied with oxygen. In this well mixed reactor, the active biomass metabolizes the dissolved and colloidal organic matter (BOD), converting it into new cellular material, carbon dioxide, and water. Following the aeration tank, the mixed liquor conceptually flows into a secondary clarifier to separate the biological floc from the treated water.

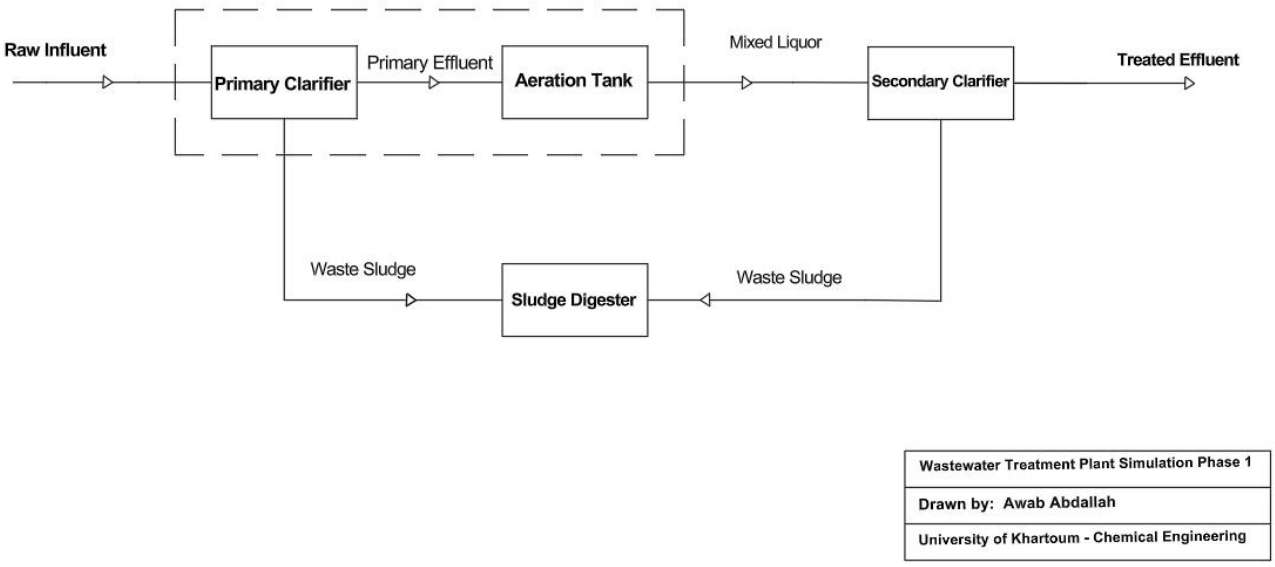

Fig. 1. Block Flow Diagram of the proposed wastewater treatment plant, illustrating the system boundaries for the Phase 1 computational simulation.

C. Sludge Handling and Processing

To handle the waste generated from the clarifiers, primary and secondary sludge streams are combined and directed to sludge processing units. The concentrated sludge undergoes biological decomposition in a digester to reduce the organic volume and generate biogas, before final dewatering and disposal.

D. *System Boundaries and Scope of the Current Simulation*

While the physical plant incorporates the comprehensive units described above, the current computational model focuses specifically on the foundational liquid phase treatment train. To establish a robust, preliminary mathematical framework, the system boundaries for the current Phase 1 simulator are constrained to the primary clarifier and the biological aeration tank (as indicated by the dashed boundary in Fig. 1).

In this simulation phase:

- **The Primary Clarifier** is modelled empirically to represent physical settling. The computational simulator utilizes a standard municipal benchmark of 30% fractional removal efficiency. This fixed efficiency rate is applied dynamically to the user's variable influent BOD input, allowing the software to calculate the exact organic load entering the bioreactor under different operating conditions.
- **The Aeration Tank** is modelled mathematically as an ideal Continuous Stirred-Tank Reactor (CSTR) governed by first-order biological decay kinetics **($k = 0.8\ hr^{-1}$).**

The complex dynamics of secondary settling, sludge recycling, and anaerobic digestion fall outside the current computational boundaries and are slated for integration in my final fifth-year expansion or the future of the project, alongside comprehensive energy balances, and overall plant cost estimation.

III. MATHEMATICAL MODELING AND MASS BALANCES

This section details the fundamental mathematical derivations used to develop the computational simulation. The equations are restricted to the unit operations defined within the current system boundary: the primary clarifier and the biological aeration tank.

The foundational mass balance framework and the first order biological kinetic assumptions utilized in this model are established upon standard chemical engineering principles [1], [2]. However, the subsequent step by step algebraic derivations specifically the isolation of the required reactor volume and the integration of the primary clarifier efficiency into a continuous algorithmic loop were custom developed to drive the logic of the computational simulator.

A. *Primary Clarifier Modeling*

In this simulation phase, the primary clarifier is modeled using an empirical performance approach rather than a strict sedimentation velocity balance. Based on standard municipal wastewater treatment benchmarks [1], a well designed primary clarifier removes approximately 30% of the influent Biochemical Oxygen Demand (BOD) associated with settleable solids.The mathematical relationship used to calculate the effluent BOD from the primary clarifier, which serves as the influent to the bioreactor (Cin,bio), is defined as:

$$C_{in,bio} = C_{in,raw}\,(1 - E_{primary})$$

Where:

- Cin,raw=Raw influent BOD concentration (mg/L)
- Eprimary = Fractional removal efficiency of the primary clarifier (0.30)

B. *Aeration Tank Biokinetics (CSTR Derivation)*

The biological aeration tank is modeled as an ideal Continuous Stirred Tank Reactor (CSTR). A macroscopic mass balance is performed across the reactor boundary.

The derivation relies on the following standard engineering assumptions:

1. The reactor operates at steady state.
2. The tank is perfectly mixed, meaning the concentration of the effluent (C_out) is identical to the concentration everywhere inside the reactor.
3. The density of the fluid remains constant, allowing the use of volumetric flow rates.
4. The biological degradation of organic matter (BOD) follows first-order decay kinetics.

*The fundamental **general mass balance** equation for a reactive system is:*

Accumulation = In – Out + Generation - Consumption

At steady state:

$$\text{Accumulation} = 0$$

Since BOD is consumed and not generated:

$$0 = FC_{in,bio} - FC_{out} - Vr_A$$

Where:

F = Volumetric flow rate (m^3/day)
V = Reactor volume (m^3)
r_A = Rate of biological consumption $(\text{mg/L} \cdot \text{day})$

First Order Reaction Kinetics:
For first order decay:

$$r_A = kC_{\text{out}}$$

Where:

$$k = \text{First-order decay constant } (\text{day}^{-1})$$

Substitution into the Mass Balance:

$$0 = FC_{\text{in,bio}} - FC_{\text{out}} - VkC_{\text{out}}$$

Rearranging:

$$FC_{\text{in,bio}} = FC_{\text{out}} + VkC_{\text{out}}$$

Factor out C_{out} *:*

$$FC_{\text{in,bio}} = C_{\text{out}}(F+Vk)$$

Dividing the equation by the volumetric flow rate (F):

$$C_{\text{in,bio}} = C_{\text{out}}\left(1+\frac{V}{F}k\right)$$

*The ratio of reactor volume to volumetric flow rate (*V/F*) is defined as* **the hydraulic retention time (HRT):**

$$\tau = \frac{V}{F}$$

Substituting:

$$C_{\text{in,bio}} = C_{\text{out}}(1+k\tau)$$

Finally, rearranging to solve for the target effluent concentration (C out), we arrive at **the governing design equation** implemented in the simulation's source code:

$$\boxed{C_{\text{out}} = \frac{C_{\text{in,bio}}}{1+k\tau}}$$

This derived equation forms **the computational core** of the **JavaScript simulation**, allowing the program to dynamically calculate effluent compliance limits based on user-defined inputs for **flow rate**, **initial concentration**, and target **retention times**.
Finally, to evaluate the total system performance against environmental regulatory standards, the overall organic Removal Efficiency (E_total) is continuously calculated by the simulator using the baseline influent and final effluent concentrations:

$$E_{\text{total}} = \left(\frac{C_{\text{in,raw}} - C_{\text{effluent}}}{C_{\text{in,raw}}}\right) \times 100$$

C. Computational Operational Modes

To power the three specific functions of the JavaScript simulator, the governing CSTR design equation was algebraically rearranged to solve for different target variables (rating mode and different design approach) based on user input.

1. Rating Mode (Evaluating an Existing Tank)

This mode calculates the final effluent concentration (*C_out*) based on a known physical reactor volume (*V*).

$$\tau = \frac{V}{F}$$

$$C_{\text{out}} = \frac{C_{\text{in, bio}}}{1+k\cdot\tau}$$

2. Design Mode: Performance Driven (Target BOD)

This mode calculates the required physical tank volume (*V req*) needed to achieve a specific, user defined environmental safety limit (*C target*). The governing equation is rearranged to solve for retention time first:

$$\tau_{\text{req}} = \frac{\left(\frac{C_{\text{in, bio}}}{C_{\text{target}}}\right) - 1}{k}$$

$$V_{\text{req}} = \tau_{\text{req}} \cdot F$$

3. Design Mode: Specification-Driven (Target HRT)

This mode calculates the necessary tank volume based on a strictly enforced residence time constraint (*τ_target*) and subsequently predicts the resulting effluent BOD.

$$V_{\text{req}} = \tau_{\text{req}} \cdot F$$

$$C_{\text{out}} = \frac{C_{\text{in, bio}}}{1+k\cdot\tau_{\text{target}}}$$

D. Computational Methodology And Simulator Development

To evaluate the mathematical models and simulate the plant's performance, a custom computational tool was developed. The development of this simulator underwent an iterative design process to align with both current academic objectives and future scalability requirements.

Iinitially, the core simulation algorithms and mass balance equations were programmed in C++. This language was selected for its high execution speed, precise memory management, and robust handling of complex numerical arrays, making it ideal for foundational engineering calculations and systems level modelling.

However, to enhance the accessibility, deployment, and real time interactivity of the simulation for this stage of the

project, the computational engine was successfully ported to a JavaScript based environment.
This transition leverages JavaScript's dynamic processing capabilities to create a highly responsive simulation tool. The current simulator allows users to instantly toggle between **"Rating" and "Design"** modes, input **custom influent parameters**, and receive immediate computational feedback on reactor sizing and compliance limits without requiring a local compiler or specific operating system dependencies.

IV. RESULTS AND DISCUSSION

A. Localized Design Basis for Simulation

To evaluate the mathematical models and demonstrate the dynamic capabilities of the computational simulator, a baseline operational scenario was established. While the greater **Khartoum** metropolitan area hosts a population in the millions, post-conflict infrastructure rehabilitation heavily favors decentralized, modular treatment facilities over vulnerable centralized systems.

To simulate a localized, neighbourhood-scale facility, the default volumetric flow rate was established at 1000 m^3/day. Assuming a conservative wastewater generation rate characteristic of water-stressed urban districts (100 liters/day per capita) [4], this baseline flow corresponds to a Population Equivalent (PE) of 10,000 residents. Furthermore, the baseline raw influent BOD was set to 310 mg/L to reflect standard medium strength domestic sewage for this demographic.

B. Computational Simulator Outputs and System Architecture Limitations

The developed JavaScript (JS) simulator operates dynamically, allowing real time evaluation of process variables. In **"Specification Driven Design" mode**, the simulator determines the required reactor volume based on a user defined target Hydraulic Retention Time (HRT) and immediately flags environmental safety statuses.

A **critical feature of the developed computational tool** is its integration of real world **physical constraints**. Standard prefabricated reactor units face highway transportation and manufacturing limits, typically capped at a maximum single unit volume of 40.0 m^3. Rather than simply outputting an impossibly large single tank, the simulator automatically triggers an architectural redesign.

For example as you can see in figure 2 , a 1000 m^3/day inflow at a 16-hour HRT demands a massive 666.67 m^3 total volume. The simulator's constraint algorithm automatically calculates that the plant must be designed utilizing 17 parallel CSTR units, each operating at a working volume of 39.22 m^3, effectively bridging the gap between theoretical mathematics and physical construction.

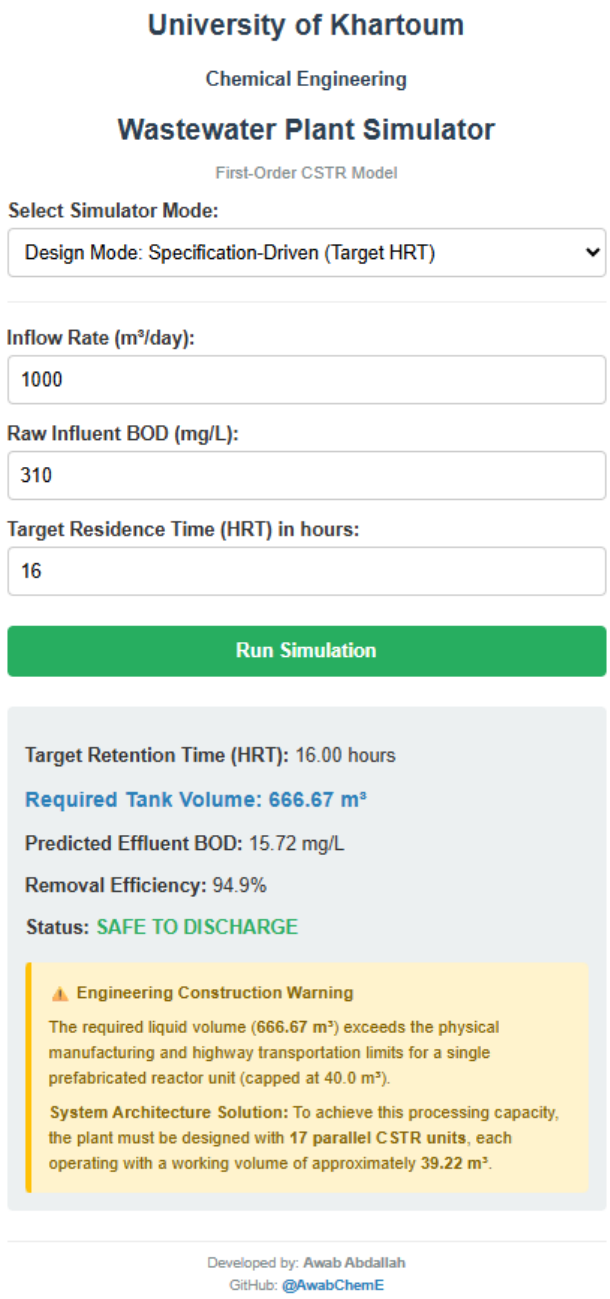

Fig. 2. Simulator output at 16-hour HRT, demonstrating automated parallel tank sizing and compliance tracking.

C. Process Validation via SuperPro Designer

To rigorously verify the accuracy of the custom JavaScript algorithm across a dynamic operating range, a multi-point validation study was conducted using SuperPro Designer, an industry-standard commercial chemical simulation software.

In SuperPro, the organic waste (BOD) was modelled utilizing Glucose as the primary substrate, and the biological degradation was driven by active Biomass. The process was simulated across three different Hydraulic Retention Times (8, 12, and 16 hours) to track the biological decay curve.

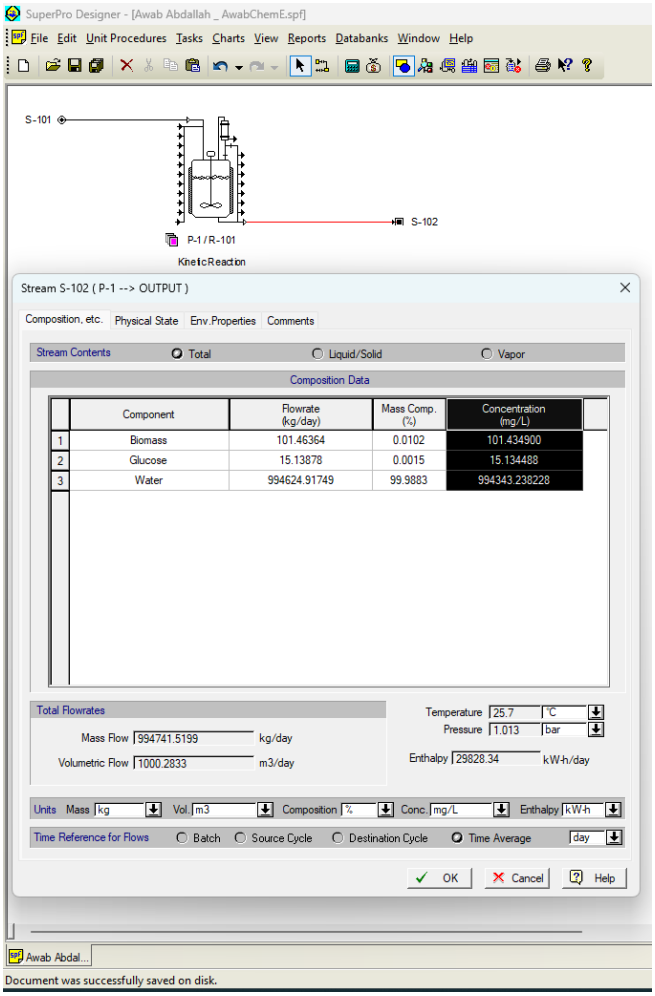

Fig. 3. SuperPro Designer process flowsheet utilized for the kinetic validation study , output at 16-hour HRT.

Table I. Algorithmic Validation: Custom JS Simulator vs. SuperPro Designer.

Target HRT (hours)	**The developed simulator Predicted Effluent BOD (mg/L)**	**SuperPro Output (Glucose) (mg/L)**	**Conservative Variance**
8	29.32	28.23	+ 3.86%
12	20.47	19.70	+ 3.90%
16	15.72	15.13	+ 3.89%

Fig. 4. The developed JavaScript Simulator vs. SuperPro Designer.

As demonstrated in Table I, the custom mathematical engine perfectly tracks the rigorous, component-based degradation kinetics of SuperPro Designer across the entire decay curve.

Notably, the JavaScript simulator exhibits a highly stable variance of less than 4%. Because the custom code slightly overestimates the final effluent concentration, it provides an inherent, conservative engineering safety margin. This ensures that a plant designed using this Phase 1 software will comfortably pass environmental regulations without risk of under sizing the reactor.

D. Future Expansion (Phase 2 Integration)

While the current engine successfully validates the concept of dynamic mass balance simulation for the primary clarifier and aeration tank, the system boundaries will be widened in the fifth year expansion phase. The computational logic will be scaled to handle the complex iterative loops required for secondary settling dynamics, sludge digester sizing, global energy balances, and total plant cost estimation.

V. CONCLUSION

This Phase 1 design report successfully demonstrates the development and validation of a dynamic computational simulator for municipal wastewater treatment. To address the need for post conflict infrastructure rehabilitation in Khartoum, the design basis was intentionally localized to model a resilient, decentralized neighborhood-scale facility processing 1000 m^3/day.

By translating the steady state, first order Continuous Stirred Tank Reactor (CSTR) mass balances into a responsive JavaScript environment, the resulting software provides real-time architectural sizing and environmental compliance tracking. The mathematical rigor of the custom algorithm was heavily validated against SuperPro Designer, yielding a conservative variance of less than 4% across a dynamic biological decay curve. Furthermore, the inclusion of physical manufacturing constraints automatically calculating the necessity of parallel reactor units bridges the gap between theoretical modeling and practical construction.

While the current JavaScript simulator serves as a highly accessible, rapid deployment proof of concept for primary clarification and aeration kinetics, the system boundaries will be significantly expanded in the fifth year phase of this project.

To handle the complex iterative loops required for secondary settling dynamics, advanced sludge digestion, global plant energy balances, and economic cost estimation, the computational architecture will be transitioned into a robust C++ environment. This foundation establishes a highly scalable framework for future municipal water management planning.

VI. AUTHOR CONTRIBUTIONS

To ensure academic transparency and establish the foundation for future project expansion, the individual contributions to this Phase 1 computational modeling and design report are outlined below:

- **Awab Abdallah:** Lead simulator architecture and algorithmic development (JavaScript), mathematical derivation of the CSTR kinetics, compilation of general municipal wastewater parameters, formal validation execution via SuperPro Designer, primary manuscript authorship, and document formatting.
- **Abdallah Mohammed:**Background literature review.
- **Mojtaba Gasmalseed:** Background literature review.

VII. ACKNOWLEDGMENT

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APPENDIX: DIGITAL SUPPLEMENT

The complete source code and executable files for the JavaScript computational simulator have been submitted directly to the advising faculty as a supplementary digital attachment.